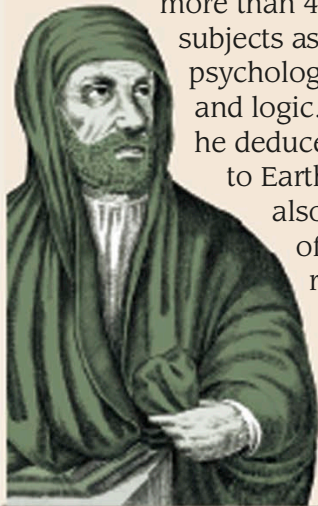


945 ▶ 1045

c 980–1037 IBN SINA

The Arab scholar Ibn Sina (also known as Avicenna) lived in Central Asia. He wrote more than 400 books on such subjects as philosophy, medicine, psychology, geology, mathematics, and logic. From direct observation, he deduced that Venus is closer to Earth than the Sun. He also developed a theory of earthquakes and their role in the formation of mountains.



Canon of Medicine

Ibn Sina's *Canon of Medicine* was one of the most important medical books in Europe and Asia during the Middle Ages. In it, he showed how Aristotle's view that there were four causes of disease could be made to agree with the theory that four humors (fluids) make up the human body.

979

Zhang Sixun's mechanical clock

Zhang Sixun, a Chinese astronomer, created an advanced mechanical clock powered by a waterwheel, which completed a full revolution every 24 hours. Every two hours, mechanical jacks emerged from inside the mechanism carrying tables that showed the time.

982

Sheng Hui Fang

Chinese physicians compiled many manuals for drug recipes during the early part of the Song Dynasty (962–1279). One of the most important of these was the *Sheng Hui Fang*, put together under government orders and containing 16,834 medicinal recipes.



945



Decimal numbers first appeared in Europe in the manuscript *Codex Vigilanus*, written by Spanish monks in 976. Knowledge of decimals had spread from the Arab world.

965

984

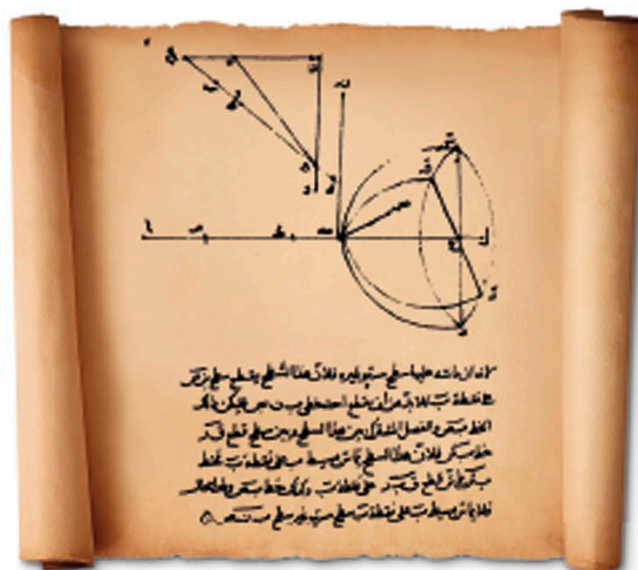
Ibn Sahl's work on refraction

Persian mathematician Ibn Sahl was interested in the refraction of light (its change in direction when it passes from one material to another). In his work *On Burning Mirrors and Lenses*, written in 984, he concluded that the amount of light that is refracted is different for each material.

Abacus in Europe

The French scholar monk Gerbert of Aurillac introduced the abacus to Europe in about 990. As a rapid way of making calculations, it was useful to astronomers, mathematicians, and merchants.

This modern abacus is very similar in design and function to the devices used 1,000 years ago.



Page from Ibn Sahl's manuscript illustrating light refraction